INCH-POUND

MS90708D
16 OCTOBER 2012
SUPERSEDING
MS90708C
28 June 1995

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL, INTERLOCKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

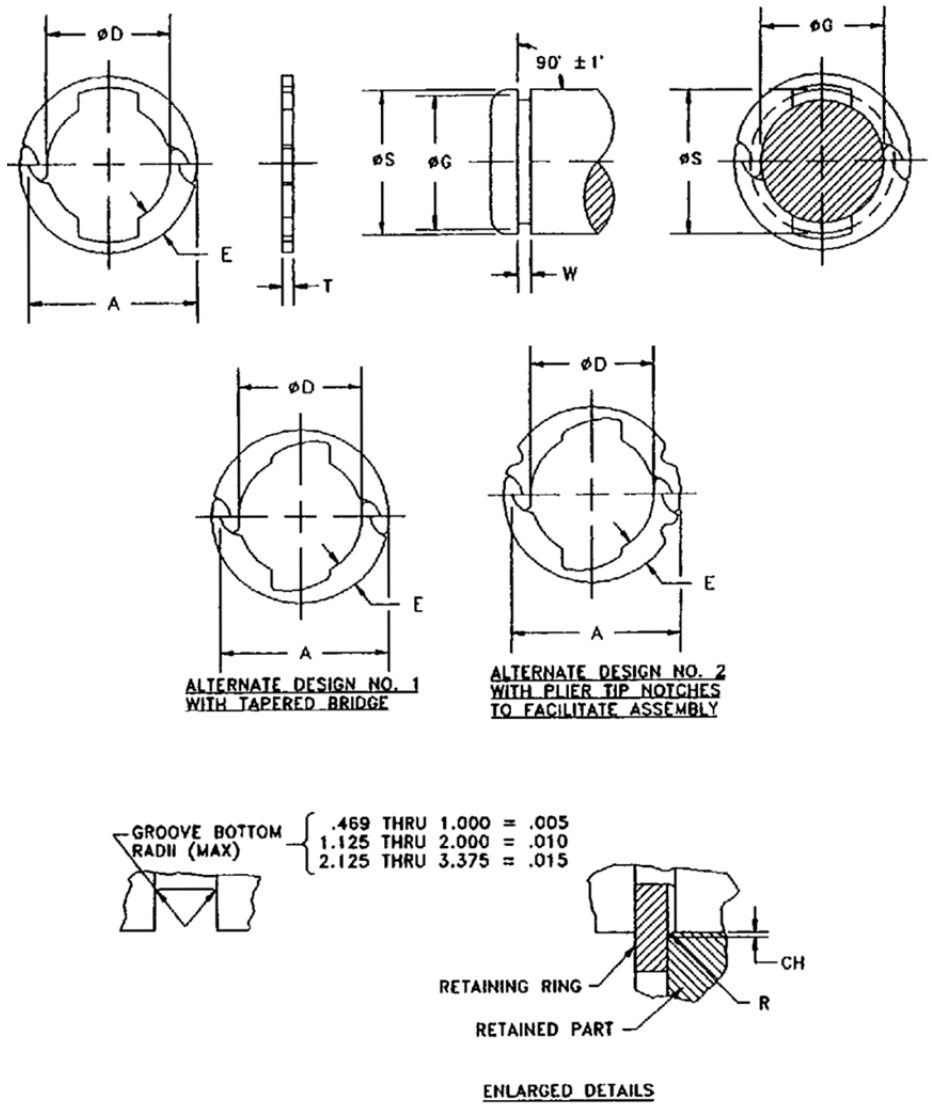


FIGURE 1. RING, RETAINING.

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TABLE 1 DIMENSIONS.

ØS SHAFT (REF)		ØD 1/ FREE (REF)	A WIDTH ACROSS LUGS		E		T 2/ THICKNESS		ØG RECOMMENDED GROOVE (REF)		W WIDTH		R 4/	CH 4/
													OF RETAINED PART (REF)	
INCH	MM	BASIC	BASIC	TOL.	BASIC	TOL.	BASIC	TOL.	BASIC	TOL.	BASIC	TOL.	MAX.	MAX.
.469	11.9	.414	.585	±.003	.105		.035		.419	±.0015	.040		.052	.040
.500	12.7	.459	.625		.105	±.005	.035		.464	.004	.040		.052	.040
.594	15.1	.538	.710	±.004	.105		.035		.544	FIM 3/	.040	+.003	.052	.040
.625	15.9	.569	.740		.105		.035		.575		.040	-.000	.052	.040
.669	17.0	.593	.815		.135		.042		.599		.047		.065	.050
				±.005				±.002		±.002				
.750	19.0	.673	.894		.135		.042		.680	.004	.047		.065	.050
.781	19.8	.703	.925		.135		.042		.711	FIM 3/	.047		.065	.050
.875	22.2	.796	1.017		.135		.042		.805		.047		.065	.050
.984	25.0	.863	1.172		.188		.050		.872	±.003	.056		.086	.066
1.000	25.4	.863	1.172	±.006	.188	±.006	.050		.872	.004 FIM 3/	.056		.081	.062
1.125	28.6	1.002	1.311		.188		.050		1.013		.056		.086	.066
1.188	30.2	1.064	1.373		.188		.050		1.075	±.003	.056		.086	.066
1.250	31.7	1.126	1.435		.188		.050		1.138	.005	.056		.086	.066
1.375	34.9	1.250	1.559	±.007	.188		.050		1.263	FIM 3/	.056	+.004	.086	.066
1.500	38.1	1.374	1.683		.188		.050		1.388		.056	-.000	.086	.066
1.562	39.7	1.412	1.778		.222		.062		1.427		.068		.100	.077
1.625	41.3	1.474	1.840		.222		.062		1.489		.068		.100	.077
1.750	44.4	1.597	1.963		.222		.062		1.614	±.005	.068		.100	.077
1.772	45.0	1.597	1.963	±.008	.222		.062		1.614	.005	.068		.094	.072
1.875	47.6	1.721	2.090		.222		.062		1.739	FIM 3/	.068		.100	.077
1.969	50.0	1.779	2.209		.262		.078		1.797		.086		.114	.088
2.000	50.8	1.809	2.242		.262	±.007	.078		1.828		.086		.114	.088
2.125	54.0	1.933	2.364		.262		.078		1.953		.086		.114	.088
2.156	54.8	1.933	2.364		.262		.078	±.003	1.953		.086		.104	.080
2.250	57.1	2.057	2.488		.262		.078		2.078		.086		.114	.088
				±.010						±.005				
2.375	60.3	2.180	2.610		.262		.078		2.203	.006	.086	+.005	.114	.088
2.500	63.5	2.304	2.734		.262		.078		2.328	FIM 3/	.086	-.000	.114	.088
2.625	66.7	2.428	2.858		.262		.078		2.453		.086		.114	.088
2.750	69.8	2.518	3.049		.323		.093		2.544		.103		.143	.110
2.875	73.0	2.642	3.172		.323		.093		2.669		.103		.143	.110
				±.012						±.006				
3.000	76.2	2.754	3.294		.329	±.008	.093		2.794	.006	.103		.143	.110
3.250	82.5	3.013	3.542		.325		.093		3.044	FIM 3/	.103		.144	.111
3.375	85.7	3.114	3.761	±.015	.395		.109		3.145		.120		.182	.140

1/ ØD = DIMENSION "D" SHOULD NOT BE USED AS PART OF AN INSPECTION PROCEDURE, USE DIMENSION "A".
2/ T = THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.
3/ FIM = (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRTICITY BETWEEN THE GROOVE AND THE SHAFT.
4/ R AND CH = RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS FOR RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS, ARE TABULATED ON PAGE 8.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 8 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A 568/A568M OR ASTM A684/A684M.

(b) CORRISION RESISTANT STEEL, IN ACCORDANCE WITH SAE AMS 5520 (UNS S15700).

(c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS

TABLE II. HARDNESS.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.469 TO .625 INCL	66.5-71.5 HR30N	44-51 HRC	56.5-62 HR30N
.669 TO 1.500 INCL	52.5-59.0 HR45N	44-51 HRC	--
1.562 TO 3.375 INCL	48.0-53.0 HRC	44-51 HRC	--
.669 TO 1.875 INCL	--	--	37-43 HRC

4. PROTECTIVE FINISH OR TREATMENT:

(a). CARBON STEEL - SHALL BE AS SPECIFIED (SEE TABLE III OR IV)

(1) CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B 696, TYPE II, CLASS 5 OR ZINC-NICKEL PLATE IN ACCORDANCE WITH SAE AMS 2417, TYPE 2.
(2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 OR ASTM B 695, TYPE II, CLASS 5.
(3) PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

(b) CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV.

EXAMPLE: MS90708-1200 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATE, EXTERNAL INTERLOCKING, RETAINING RING FOR USE ON A 2.000 DIAMETER SHAFT.

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TABLE III. DASH NUMBERS FOR MS90708

Ø S SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE OR ZINC-NICKLE PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT	STEEL, CORROSION RESISTANT	1/ BERYLLIUM COPPER
	DASH NO	DASH NO	DASH NO	DASH NO	DASH NO
.469	-1046	-2046	-3046	-4046	-5046
.500	-1050	-2050	-3050	-4050	-5050
.594	-1059	-2059	-3059	-4059	-5059
.625	-1062	-2062	-3062	-4062	-5062
.669	-1066	-2066	-3066	-4066	-5066
.750	-1075	-2075	-3075	-4075	-5075
.781	-1078	-2078	-3078	-4078	-5078
.875	-1087	-2087	-3087	-4087	-5087
.984	-1098	-2098	-3098	-4098	-5098
1.000	-1098	-2098	-3098	-4098	-5098
1.125	-1112	-2112	-3112	-4112	-5112
1.188	-1118	-2118	-3118	-4118	-5118
1.250	-1125	-2125	-3125	-4125	-5125
1.375	-1137	-2137	-3137	-4137	-5137
1.500	-1150	-2150	-3150	-4150	-5150
1.562	-1156	-2156	-3156	-4156	-5156
1.625	-1162	-2162	-3162	-4162	-5162
1.750	-1175	-2175	-3175	-4175	-5175
1.772	-1175	-2175	-3175	-4175	-5175
1.875	-1187	-2187	-3187	-4187	-5187
1.969	-1196	-2196	-3196	-4196	
2.000	-1200	-2200	-3200	-4200	
2.125	-1212	-2212	-3212	-4212	
2.156	-1212	-2212	-3212	-4212	
2.250	-1225	-2225	-3225	-4225	
2.375	-1237	-2237	-3237	-4237	
2.500	-1250	-2250	-3250	-4250	
2.625	-1262	-2262	-3262	-4262	
2.750	-1275	-2275	-3275	-4275	
2.875	-1287	-2287	-3287	-4287	
3.000	-1300	-2300	-3300	-4300	
3.250	-1325	-2325	-3325	-4325	
3.375	-1337	-2337	-3337	-4337	

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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TABLE IV. SUBSTITUTION TABLE, CROSS REFERENCE OF PART NUMBERS

Ø S SHAFT (REF)	INACTIVE	SUBSTITUTE	SUBSTITUTE	SUBSTITUTE
	CARBON STEEL	1/ CARBON STEEL CADMIUM PLATE OR ZINC-NICKLE PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT
	MS90708	MS90708	MS90708	MS90708
.469	-46	-1046	-2046	-3046
.500	-50	-1050	-2050	-3050
.594	-59	-1059	-2059	-3059
.625	-62	-1062	-2062	-3062
.669	-66	-1066	-2066	-3066
.750	-75	-1075	-2075	-3075
.781	-78	-1078	-2078	-3078
.875	-87	-1087	-2087	-3087
.984	-98	-1098	-2098	-3098
1.000	-98	-1098	-2098	-3098
1.125	-112	-1112	-2112	-3112
1.188	-118	-1118	-2118	-3118
1.250	-125	-1125	-2125	-3125
1.375	-137	-1137	-2137	-3137
1.500	-150	-1150	-2150	-3150
1.562	-156	-1156	-2156	-3156
1.625	-162	-1162	-2162	-3162
1.750	-175	-1175	-2175	-3175
1.772	-175	-1175	-2175	-3175
1.875	-187	-1187	-2187	-3187
1.969	-196	-1196	-2196	-3196
2.000	-200	-1200	-2200	-3200
2.125	-212	-1212	-2212	-3212
2.156	-212	-1212	-2212	-3212
2.250	-225	-1225	-2225	-3225
2.375	-237	-1237	-2237	-3237
2.500	-250	-1250	-2250	-3250
2.625	-262	-1262	-2262	-3262
2.750	-275	-1275	-2275	-3275
2.875	-287	-1287	-2287	-3287
3.000	-300	-1300	-2300	-3300
3.250	-325	-1325	-2325	-3325
3.375	-337	-1337	-2337	-3337

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. UNLESS OTHERWISE SPECIFIED, ISSUES OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICATION.

4. RECOMMENDED DESIGN LIMITATIONS AND USAGE.

(a) INTENDED USE- TO PROVIDE HIGH CIRCULAR SHOULDERS FOR POSITIONING AND RETAINING MACHINE COMPONENTS ON SHAFTS. THE IDENTICAL SEMI-CIRCULAR HALVES HELD TOGETHER BY THE INTERLOCKING PRONGS FORM A BALLANCED RING CONCENTRIC WITH THE SHAFT, WHICH WILL WITHSTAND HIGH ROTATIONAL SPEEDS. THE USE OF THE FOLLOWING FORMULAS IS BASED ON THE FACT THAT THE MATERIAL WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE- THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RING HALVES SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED. (PROVIDING GROOVE HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

TABLE V. APPROXIMATE SAFETY RPM LIMITS.

Ø SHAFT(INCHES)		.500	1.000	1.500	2.000	2.500	3.000	3.375
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	50,000	30,000	22,000	15,000	12,000	9,000	6,800
BERYLLIUM COPPER	RPM LIMIT	32,000	19,000	14,000				

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$$P = \frac{C_F \pi S T X}{F}$$

WHERE:

P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (P.S.I.) 1/
F = FACTOR OF SAFETY, F=3 IS RECOMMENDED TO INSURE A SAFE WORKING LOAD.
C_F = CONVERSION FACTOR, C_F = 3/4 IS RECOMMENDED SINCE SHEAR AREA IS REDUCED AT THE CUTOUTS IN THE RING.

1/ X = 150,000 P.S.I. ULTIMATE SHEAR STRENGTH FOR RINGS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 P.S.I. ULTIMATE SHEAR STRENGTH FOR RINGS OF BERYLLIUM COPPER.

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(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL

$$P = \frac{C_F \pi S d Y}{F}$$

WHERE:

P = ALLOWABLE COMPRESSION LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY, F= 2 IS RECOMMENDED TO INSURE A SAFE WORKING LOAD.
C_F = CONVERSION FACTOR, C_F = 3/4 IS RECOMMENDED SINCE CONTACT AREA IN THE GROOVE IS REDUCED AT THE CUTOUTS IN THE RING

(d) THE MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$$Z = 3d$$

WHERE:

Z = MINIMUM DISTANCE BEWTWEEN OUTER GROOVE WALL AND END OF SHAFT (INCHES).
d = GROOVE DEPTH (INCHES).

(e) ALLOWABLE SHAFT DIAMETER =

$$S = \sqrt{G^2 + \frac{4FP}{Y \pi C_F}}$$

WHERE:

S = ALLOWABLE SHAFT DIAMETER
G = GROOVE DIAMETER.
F = FACTOR OF SAFETY, SEE FORMULA (c) ABOVE.
P = DESIGN LOAD.
Y = YIELD STRENGTH IN COMPRESSION OF GROOVE MATERIAL (P.S.I.).
C_F CONVERSION FACTOR, SEE FORMULA (c) ABOVE

(f) DIFFERENTIAL ROTATION =

DIFFERENTIAL ROTATION OF RING AND ADJACENT PART CREATES NO ELEMENT OF RISK IN THE APPLICATION OF THE RING UP TO THE ALLOWABLE LOAD CALCULATED TO THE ABOVE FORMULA (b).

(g) IMPACT CAPACITY OF RING OR GROOVE WALL =

$$I_R = \frac{PT}{2}$$ FOR THE RING (INCH POUNDS) ABUTTING COMPONENTS TO HAVE SHARP CORNERS

$$I_G = \frac{Pd}{2}$$ FOR THE GROOVE (INCH POUNDS).

WHERE:

P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS).
T = RING THICKNESS (INCHES).
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS).
d = GROOVE DEPTH (INCHES).
I_R = IMPACT CAPACITY OF RING (INCH POUNDS).

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(h) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED:
WHEN THE RADIUS OR CHAMFER OF THE RETAINED PART DOES NOT EXCEED THE MAXIMUM RADIUS ALLOWED FOR THE BOTTOM OF THE RING GROOVE, THE LESSER LOAD CAPACITY COMPUTED FROM THE FORMULAS ON PAGE 4 WILL APPLY. THE CORNER RADII AND CHAMFERS LISTED ON PAGE 2 WERE CHOSEN AS LARGE AS POSSIBLE FOR THE RING SIZES INVOLVED AND ARE RELATED TO THE MAXIMUM THRUST LOADS LISTED IN TABLE VI.

IF THE CORNER RADII OR CHAMFERS ARE SMALLER THAN THOSE LISTED, THEN THE THRUST LOADS INCREASE PROPORTIONALLY IN ACCORDANCE WITH THE FOLLOWING FORMULAS:

$P^1 = \frac{PCH}{CH^1}$ OR $P^1 = \frac{PR}{R^1}$

WHERE:

P^1 = NEW ALLOWABLE THRUST LOAD.
P = LISTED ALLOWABLE THRUST LOAD.
CH^1 = NEW (SMALLER) CHAMFER.
CH = LISTED CHAMFER.
R^1 = NEW (SMALLER) CORNER RADIUS.
R = LISTED CORNER RADIUS.

LIMIT LOADS LISTED BELOW ARE BASED ON RINGS OF STEEL (WORKING STRESS = 250,000 P.S.I.) AND OF BERYLLIUM COPPER (WORKING STRESS = 180,000 P.S.I.). IF THE ALLOWABLE GROOVE CAPACITY LOADS AS CALCULATED BY USING THE FORMULA GIVEN ABOVE ARE LESS, THEN THEY SHOULD BE USED.
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TABLE VI. LIMIT LOADS.

NOMINAL RING SIZE		ALLOWABLE THRUST LOAD FOR RING ASSEMBLIES WITH PARTS HAVING MAXIMUM CORNER RADII OR CHAMFERS	
FROM	TO	CARBON STEEL OR CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.496	.625	610 LB	440 LB
.669	.875	880 LB	630 LB
.984	1.500	1250 LB	900 LB
1.562	1.875	1900 LB	1370 LB
1.969	2.625	3050 LB	--
2.750	3.250	4300 LB	--
3.375	--	5950 LB	--

5. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

MILITARY INTEREST

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-018)

Review activities:
Army – AT, AV, CR, CR4
Navy – MC, SH, YD
Air Force - 71

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